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Targeting GALNT4 suppresses atherosclerosis via TNFR1/NF-κB mediated attenuation of endothelial inflammation

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ABSTRACT

Endothelial inflammation is a critical driver of atherosclerosis (AS). Although *N*-acetylgalactosaminyltransferase 4 (GALNT4) has been genetically linked to coronary artery disease, its specific function and mechanism within endothelial cells (ECs) during AS progression remain poorly understood, presenting a significant knowledge gap. This study aimed to elucidate the precise role of GALNT4 in endothelial inflammation and AS development. GALNT4 expression was assessed in human and mouse atherosclerotic plaques. *ApoE*$^{-/-}$ mice fed on a high-fat diet (HFD) received AAV-shRNA-mediated GALNT4 knockdown to evaluate atherosclerotic lesions and vascular inflammation. In vitro, gain and loss-of-function studies were conducted in TNF-α-treated human umbilical vein endothelial cells (HUVECs) to investigate monocyte adhesion and NF-κB activation. Mechanistic insights were gained through lectin blot, co-immunoprecipitation, and chromatin immunoprecipitation (ChIP) assays. Results showed that GALNT4 expression was significantly upregulated in human and mouse atherosclerotic plaques. In *ApoE*$^{-/-}$ mice, GALNT4 knockdown markedly attenuated atherosclerotic lesion area (45.1 % reduction) and reduced ICAM-1/VCAM-1 expression without altering plasma lipids. In HUVECs, TNF-α induced GALNT4 expression in a dose and time-dependent manner. Functionally, GALNT4 knockdown suppressed TNF-α-induced monocyte adhesion, while its overexpression exacerbated these effects. Mechanistically, GALNT4 knockdown reduced O-GalNAcylation of TNFR1, diminished TNF-α-triggered phosphorylation and nuclear translocation of NF-κB p65, and inhibited NF-κB binding to the ICAM-1 and VCAM-1 promoters. Our findings demonstrate that GALNT4, upregulated in AS, promotes endothelial inflammation by enhancing the O-GalNAcylation of TNFR1 and subsequent activation of the NF-κB pathway. Targeting GALNT4 represents a promising therapeutic strategy for attenuating AS.

1. Introduction

Atherosclerosis (AS) is the leading cause of most cardiovascular diseases, resulting in a high rate of mortality worldwide [1,2]. Endothelial dysfunction and vascular inflammation play essential roles in the initiation and progression of AS [3]. Proinflammatory stimuli, such as tumor necrosis factor-α (TNF-α), induce the expression of cell adhesion molecules in endothelial cells (ECs) [4]. Intercellular adhesion molecule 1 (ICAM-1) and vascular cell adhesion molecule 1 (VCAM-1) mediate inflammation and promote adherence of leukocytes to ECs [5,6].

N-acetylgalactosaminyltransferase 4 (GALNT4) is a member of the GalNAc-T family responsible for initiating the attachment of the first *N*-acetylgalactosamine (GalNAc) monosaccharide to target proteins during O-linked glycosylation [7,8]. It is located in the 12q21.33 region, which has been linked to coronary artery disease in Chinese Han ancestry [9]. Previous research demonstrated that the lead SNP, rs7136259, located in the 12q21.33 locus, is related to the expression of GALNT4. GALNT4 is upregulated in intimal lesions, vascular smooth muscle cells(SMCs), and diseased aortic ECs of the *ApoE*$^{-/-}$ mice, but downregulated in foam cells [10], suggesting a multi-cellular role for this gene. More recent

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evidence indicates that GALNT4 O-glycosylates threonine (Thr) residues of the P-selectin glycoprotein ligand (PSGL-1), which is crucial in endothelial-platelet interactions in monocytes [11]. GALNT4 may be a potential therapeutic target for AS. However, the role and mechanisms of GALNT4 in ECs remain poorly understood.

In this study, we investigated the novel function of GALNT4 in the progression of AS, focusing on its regulation of endothelial inflammation.

2. Materials and methods

2.1. Human subjects

Atherosclerotic arterial samples were collected from patients (aged 40–80 years) undergoing abdominal aorta surgery or carotid endarterectomy at the First Affiliated Hospital of Xinxiang Medical University, all with imaging-confirmed lesions and without active infections, autoimmune diseases, malignancy, or severe hepatic/renal dysfunction. Normal arterial tissues were obtained from road traffic accident fatalities within the same age range, who exhibited no macroscopic atherosclerosis, no history of cardiovascular disease or diabetes, and no significant aortic trauma. All studies involving humans complied with the Declaration of Helsinki and received approval from the Ethics Committee of Xinxiang Medical University (Approval No. XXMU2022-K0157). Written informed consent was obtained from all participants or their relatives in the case of deceased individuals before surgery.

2.2. Animals

Eight-week-old Male *ApoE*$^{-/-}$ mice and a high-fat diet (HFD, D1208C, 0.15 % cholesterol and 21 % fat) were purchased from SiPeiFu Company (Beijing, China). The recombinant adeno-associated virus (serotype 9) containing short hairpin RNA targeting GALNT4 (AAV-shGALNT4) and the negative control shRNA (AAV-shNC) was designed by Hanbio Biotechnology (Shanghai, China).

The *ApoE*$^{-/-}$ mice were randomly divided into 4 groups and given ad libitum water access while being kept on a 12-h light-dark cycle. The groups were: (1) Control group: *ApoE*$^{-/-}$mice fed a chow diet for 12 weeks. (2) AS group: *ApoE*$^{-/-}$mice fed an HFD for 12 weeks. (3) AAV-shNC group: *ApoE*$^{-/-}$mice fed an HFD and treated with AAV-shNC. These mice received injections of 5×10^8 plaque-forming units (pfu) of AAV-shNC via the tail vein for 2 weeks before feeding an HFD. (4) AAV-shGALNT4 group: *ApoE*$^{-/-}$mice fed HFD and treated with AAV-shGALNT4. Similar to the AAV-shNC group, these mice received injections of 5×10^8 pfu of AAV-shGALNT4 via the tail vein for 2 weeks before feeding an HFD. All animal experiments adhered to the NIH Animal Care and Use Committee guidelines and were approved by the Care of Experimental Animals Committee of Xinxiang Medical University (Approval No. XYLL-20210116).

2.3. Atherosclerotic lesion analysis

The mouse aorta was collected from the base of the ascending aorta to the iliac bifurcation to assess atherosclerotic lesions. The entire aorta was carefully dissected, and the atherosclerotic lesions were stained with oil red O (ORO) (Servicebio, Wuhan, China), followed by quantification through surface analysis.

Aortic root analysis was performed as previously described [12]. Briefly, the tissue was embedded in optimal cutting temperature compound (OCT) and serially sectioned at 10 μm thickness. Five consecutive sections per mouse, spaced 200 μm apart, were collected starting from the aortic valve leaflets and covering the aortic sinus. Lesion areas were quantified using ORO staining and averaged across the five sections to determine the atherosclerotic burden. Measurements were performed using ImageJ software (NIH). Additionally, human arterial tissue and mouse aortic roots were fixed in 10 % buffered neutral formalin, followed by embedding in paraffin or OCT compound. The sections, measuring 5 μm, were dewaxed, rehydrated, and stained with hematoxylin and eosin (H&E) to evaluate the damage caused by the lesions. The area of the atherosclerotic lesions was quantified using Image J software (NIH).

2.4. Immunofluorescent (IF) staining

Human and mouse aorta were isolated and fixed with 4 % neutralized paraformaldehyde for 15 min. After being blocked by 1 % BSA for 30 min at RT, aortas were incubated in 1 % BSA with antibody GALNT4 and CD31 (Abcam, USA) overnight at 4 °C. After being washed in PBS 3 times, the aortas were incubated with Alexa-Fluor 488- or Alexa-Fluor 647-conjugated secondary antibodies(Proteintech, China) for 1 h at RT. The fluorescent signals were detected by a fluorescence microscope (Nikon, Japan).

2.5. Cell culture

As previously reported, primary human umbilical vein endothelial cells (HUVECs) were isolated from the human umbilical vein vascular wall [13]. The cells were then cultured in an endothelial cell culture medium (ECM) (ScienCell, USA) according to the manufacturer's instructions. For the experiments, cells from passages 3–6 were used. HUVECs were seeded onto a 6- or 12-well dish and treated with varying concentrations of TNF-α (0, 10, 20, or 30 ng/mL) or with 20 ng/mL of TNF-α for different durations (0, 6, 12, 24, or 48 h).

THP-1 cells were acquired from the American Type Culture Collection (ATCC, USA) and cultured in RPMI-1640 complete medium (10 % FBS).

2.6. Adenovirus construction

The recombinant adenovirus (AdGALNT4) was designed by Hanbio Biotechnology (Shanghai, China) to overexpress the full-length human GALNT4 cDNA. The empty vector control (AdE) contained the same backbone and promoter but lacked the GALNT4 insert.

2.7. siRNA transfection

Human GALNT4 siRNA, human p65 siRNA, and negative control (NC) siRNAs were synthesized by GenePharma Biotechnology (Shanghai, China). The siRNA sequences are listed in Supplementary **Table I**. siRNAs were transfected into cells using Lipofectamine RNAiMAX (Invitrogen, USA) following the manufacturer's instructions.

2.8. RNA isolation and qRT-PCR

Total RNA from human plaques, mouse tissues, or HUVECs was isolated using a TRIzol reagent (Invitrogen, USA). cDNA was obtained using HiScript II 1st Strand cDNA Synthesis Kit (Vazyme, Nanjing, China), and qPCRs were performed using ChamQ SYBR qPCR Master Mix (Vazyme, Nanjing, China). GAPDH was used for normalization. Primer sequences used in this study are listed in Supplementary **Table II**.

2.9. Western blot and lectin blot analysis

Total protein from the entire aorta and HUVECs was extracted using RIPA cell/tissue lysis buffer (Beyotime, Shanghai, China) following the manufacturer's instructions. Equal amounts of proteins were electrophoresed following previously established protocols [14]. The primary antibodies used include GALNT4, VCAM-1, ICAM-1, TNFR1, phospho-NF-κB p65, and NF-κB p65 were purchased from Affinity Biosciences (Cincinnati, USA). β-actin, Goat anti-rabbit, and anti-mouse secondary antibodies were purchased from Wanleibio (Shenyang, China). For the lectin blot, the membranes were incubated with Biotinylated *Vicia*

Villosa Lectin (1:8000, VECTOR Laboratories, USA) for 12 h and incubated with horseradish peroxidase-labeled streptavidin (1:8000, ZSGB-BIO, China) at room temperature for 1 h.

Image-Pro Plus 6.0 software was used to analyze the protein gray value and obtain the quantitative relative protein density.

2.10. Lectin pull-down and immunoprecipitation

Cell lysates were incubated with agarose-conjugated *Vicia villosa* lectin (VVL; 1 μg/mL; VECTOR Laboratories, USA) for 12 h at 4 °C. Immunoprecipitation was performed using a commercial kit (Beyotime, Shanghai, China) according to the manufacturer's protocol. Briefly, cell lysates were incubated overnight at 4 °C with an anti-TNFR1 antibody (Proteintech, Wuhan, China; diluted 1:100). Subsequently, 50 μL of Protein G-agarose beads were added, and the mixture was rotated end-over-end overnight at 4 °C. Precipitated complexes from both assays were then analyzed by Western blotting and lectin blotting as described in the respective protocols.

2.11. Monocyte-endothelial cell adhesion assay

THP-1 cells were labeled by incubating with Cell Tracker™ CM-Dil (Invitrogen, USA) for 20 min at 37 °C. The labeled THP-1 cells were then added to monolayers of HUVECs and incubated for 30 min. After this incubation period, non-adherent THP-1 cells were removed. Finally, we captured the images and counted the number of labeled cells from 6 random fields using a fluorescence microscope (Nikon, Japan). Three independent biological replicates were performed, each involving HUVECs isolated from distinct donors. For each experimental condition, three technical replicate wells were analyzed, and six random fields per well were imaged. Data from the six fields within a well were averaged to generate a single value per technical replicate.

2.12. NF-κB translocation

According to the manufacturer's instructions, NF-κB activation was detected by the Nuclear Translocation Assay Kit (Beyotime, Shanghai, China). Briefly, HUVECs were fixed for 5–15 min. Then the cells were incubated in an immunostaining sealing solution for 1 h at room temperature before being exposed to a 1:200 dilution of anti-NF-κB-p65 antibody overnight at 4 °C. Afterward, the cells were washed and stained with anti-mouse Cy3 and incubated for 1 h at room temperature, and the cell nuclei were stained with DAPI for 5 min. The images were captured using a Nikon fluorescence microscope.

2.13. ChIP assay

Chromatin immunoprecipitation (ChIP) assays were conducted using the BeyoChIP™ Enzymatic ChIP Assay Kit (Beyotime, Shanghai, China) as previously reported [15]. DNA-bound NF-κB subunit was immunoprecipitated by incubating with an antibody against P65 and an IgG antibody(used as a negative control) overnight at 4 °C. Next, the crosslinking of protein-DNA complexes was reversed, and DNA was purified using spin columns following the manufacturer's instructions. The purified DNA was amplified via PCR to identify the promoter region containing NF-κB binding. The primers are listed in Supplementary **Table II**.

2.14. Statistical analysis

SPSS 20.0 and GraphPad Prism software were used to perform statistical analysis. For comparisons between two groups of normally distributed data, we used Student's *t*-test if the variances were similar, or Welch's correction if equal variances were not assumed, as determined by an F-test. For comparisons involving more than two groups, the Brown-Forsythe test was used to assess similar variances, followed by a standard ANOVA or the Welch ANOVA, depending on whether similar variances were assumed or not. A *P*-value of less than 0.05 was considered statistically significant.

3. Results

3.1. GALNT4 was upregulated in human and mouse atherosclerotic plaques

To determine the relationship between GALNT4 and AS, we first assessed GALNT4 expression in human atherosclerotic arteries. The baseline characteristics and serum lipid levels of normal subjects and those with AS were presented in Supplementary Table III. H&E staining of the aortas of AS patients demonstrated that the normal aortic intima was intact and smooth. In contrast, atherosclerotic samples exhibited intimal thickening, disorganized muscle fibers, and a loose tissue structure (Fig. 1A). By Western blot, we found significantly higher GALNT4 levels in atherosclerotic plaques compared to normal arterial tissues (Fig. 1B and C). Since endothelial dysfunction is a well-established early event in AS pathogenesis, we further examined the endothelial expression of GALNT4 in atherosclerotic plaques by performing a dual immunofluorescent assay using antibodies against GALNT4 and CD31 (a marker of endothelial cells). IF staining demonstrated that approximately 57.6 % of CD31+ cells (red) were also positively stained with GALNT4 (green) in human atherosclerotic plaques (Fig. 1D). Quantitative analysis using Manders' overlap coefficient indicated a 3.58-fold increase in GALNT4-CD31 co-localization in human atherosclerotic plaques compared to normal arteries ($P < 0.01$, Fig. 1E).

These findings were also confirmed using mouse models. H&E staining and ORO staining of the aortas of AS mice were shown in Fig. 1F. Consistent with the human data, GALNT4 expression was significantly upregulated in mouse atherosclerotic plaques (Fig. 1G and H). IF staining further demonstrated a 2.63-fold increase in GALNT4 expression in ECs of mouse aortic plaques relative to normal arteries ($P < 0.05$, Fig. 1I and J). Although endothelial cell loss is a hallmark of advanced AS, the remaining CD31-positive ECs overlying atherosclerotic plaques exhibited markedly enhanced GALNT4 expression, suggesting a reactive upregulation in the activated endothelium.

Collectively, these findings indicated that GALNT4 expression is higher in ECs of atherosclerotic plaques than in those of normal aortas, suggesting that ECs contribute to the upregulation of GALNT4 during the progression of AS.

3.2. Targeting GALNT4 repressed atherosclerosis formation and endothelial inflammation in vivo

To determine the functional role of elevated GALNT4 in AS, we knocked down GALNT4 in *ApoE*$^{-/-}$ mice fed HFD using AAV-shGALNT4. First, we administered AAV9-GFP (the same serotype as AAV-shGALNT4) via tail vein injection in *ApoE*$^{-/-}$ mice. Fluorescence imaging of aortic sections confirmed GFP expression in the aortic endothelium, indicating successful viral delivery to the vessel wall (Supplementary Fig. S1A). Western blot analysis demonstrated efficient GALNT4 knockdown in the murine aorta, with a 87.1 % reduction in protein levels (Fig. 2A and B). At the same time, we performed double IF staining for CD31 and GALNT4, along with immunohistochemistry (IHC) for GALNT4 on aortic sections from AAV-shGALNT4-treated mice. The results revealed a marked decrease in the mean fluorescence intensity of GALNT4 within CD31+ ECs of the AAV-shGALNT4 group compared to the AAV-shNC control group (Supplementary Fig. S1B to E). Furthermore, Western blot analysis of CD31 and key apoptosis-related proteins (Bax and Cleaved-caspase-3) in the mouse aorta showed no significant differences between the two groups (Supplementary Fig. S2A and B). These data confirm that AAV-shGALNT4 treatment specifically reduces GALNT4 expression without

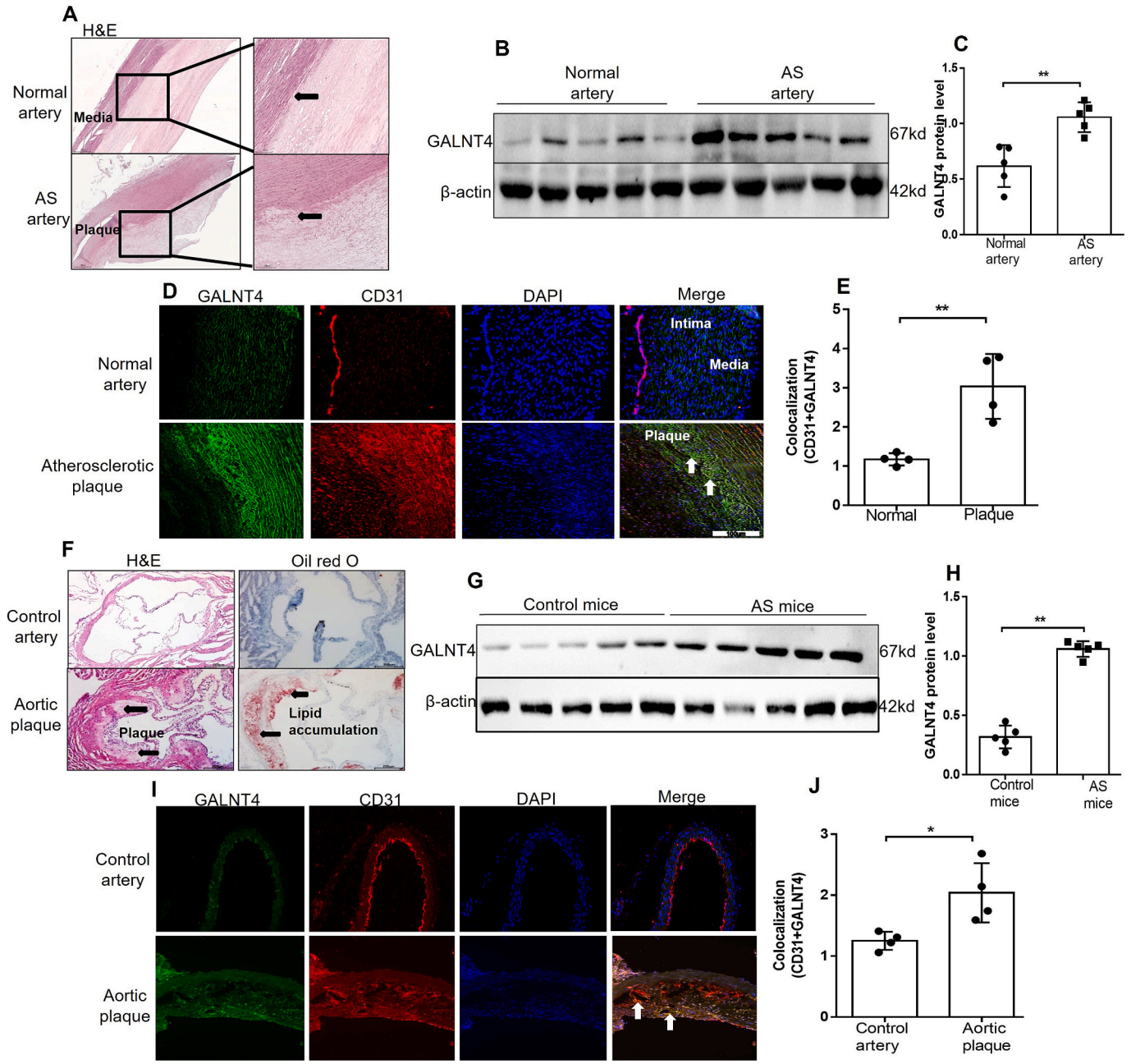


Fig. 1. GALNT4 was upregulated in human and mouse atherosclerotic plaques.
(A) Representative H&E staining of human normal and atherosclerotic arteries. (B and C) Western blot analysis of GALNT4 protein level (relative to the internal reference protein β-actin) in human normal and atherosclerotic arteries. (D) Representative IF staining for GALNT4 (green) and CD31 (red) in human normal aortas and atherosclerotic arteries. DAPI (blue) stains the nuclei. Scale bar: 100 μm. (E) The percentages of GALNT4 + $CD31^+$ cells relative to the total $CD31^+$ cells were quantified from tissue sections in D($n = 3$). (F) Representative H&E and oil red O staining of aortic root sections in normal aortas and aortic plaques in mice. (G and H) Western blot analysis of GALNT4 protein level (relative to the internal reference protein β-actin) in normal aortas and aortic plaques of mice. (I) Representative IF staining for GALNT4 (green) and CD31 (red) in normal aortas and aortic plaques in mice($n = 4$). Scale bar: 100 μm. (G) The percentages of GALNT4 + $CD31^+$ cells relative to the total $CD31^+$ cells were quantified from tissue sections in F(n = 4). Normal: Normal arteries. AS: Atherosclerotic arteries. $^*P < 0.05$, $^{**}P < 0.01$ versus Normal, tested by the two-tailed Student's *t*-test. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

compromising endothelial cell integrity.

Then, en-face oil red O (ORO) staining showed significantly reduced lipid accumulation (indicated by red staining) in the aortas of the AAV-shGALNT4 group relative to the AAV-shNC group(Fig. 2C and D). Consistent with this, H&E staining (Fig. 2E) and ORO staining of the aortic roots showed that the lesion area at the aortic sinus decreased by 45.1 % with GALNT4 knockdown (Fig. 2F and G). Additionally, GALNT4 knockdown resulted in reduced protein levels of VCAM-1 and ICAM-1 in $ApoE^{-/-}$ mice (Fig. 2H and I). We also measured the plasma lipid parameters (total cholesterol [TC], triglycerides [TG], low-density lipoprotein cholesterol [LDL-C], and high-density lipoprotein cholesterol [HDL—C]) in $ApoE^{-/-}$ mice injected with AAV-shNC or AAV-shGALNT4. No significant differences were observed in plasma TC, TG, LDL-C, or HDL-C levels between the AAV-shGALNT4 and AAV-shNC groups (all $P > 0.05$, Supplementary Fig.S3).

These results confirm that GALNT4 knockdown does not alter

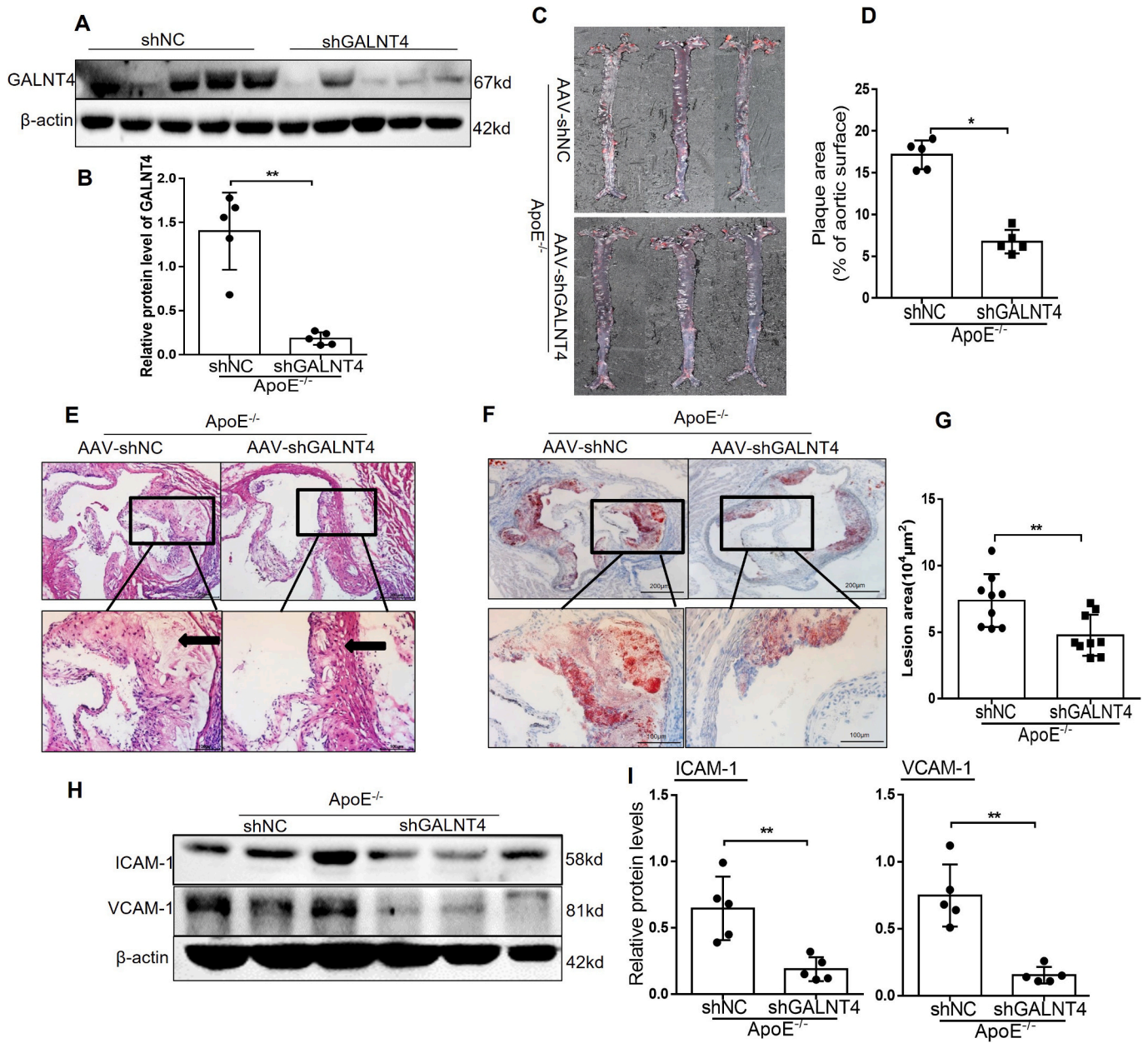


Fig. 2. Targeting GALNT4 attenuated endothelial dysfunction and atherosclerosis in *ApoE*$^{-/-}$ mice fed the high-fat diet.
(A and B) Western blot analysis of GALNT4 protein level in the vessel of mice injected with AAV-shNC or AAV-shGALNT4. (C) Representative *en face* oil red O (ORO-red) staining in aortas of *ApoE*$^{-/-}$ mice injected with AAV-shNC or AAV-shGALNT4. (D) Quantifying plaque areas as a percentage of total aortic areas. (E) Representative H&E staining of aortic root sections from *ApoE*$^{-/-}$ mice injected with AAV-shNC or AAV-shGALNT4. (F) Representative images of ORO-stained aortic roots of *ApoE*$^{-/-}$ mice injected with AAV-shNC or AAV-shGALNT4. (G) Quantification of aortic root lesion areas($n = 9$). (H) Western blot analysis of protein levels of ICAM-1, VCAM-1 in the vessel tissues of *ApoE*$^{-/-}$ mice injected with AAV-shNC or AAV-shGALNT4. (I) Quantification of protein levels of ICAM-1, VCAM-1. *$P < 0.05$, **$P < 0.01$ versus AAV-shNC group, tested by two-tailed Student's *t*-test. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

systemic lipid metabolism in this model. Therefore, the observed reduction in atherosclerotic lesions is likely attributable to its anti-inflammatory effects on ECs rather than to lipid-lowering mechanisms.

3.3. GALNT4 was induced in HUVECs along with changes in the expression of adhesion molecules

Since the inflammatory microenvironment of atherosclerotic plaques in HFD-fed mice is rich in cytokines, including TNF-α [16], we next examined whether the pro-inflammatory cytokine TNF-α induces GALNT4 expression in cultured HUVECs. Our results demonstrated that GALNT4 expression was significantly elevated at TNF-α concentrations of 20 ng/mL or higher. Correspondingly, ICAM-1 and VCAM-1 levels were also considerably increased following TNF-α treatment (Fig. 3A and B). Subsequently, HUVECs were treated with TNF-α for different durations. After 6 h of TNF-α stimulation, both mRNA and protein levels of GALNT4 were significantly increased compared to the 0 h control ($P < 0.05$). In contrast, the upregulation of ICAM-1 and VCAM-1 at this early time point did not reach statistical significance. GALNT4 expression peaked at 12 h post-stimulation, as did the expression of ICAM-1 and VCAM-1, indicating a similar temporal pattern among these molecules (Fig. 3C to E). Similar to TNF-α, other inflammatory stimuli such as

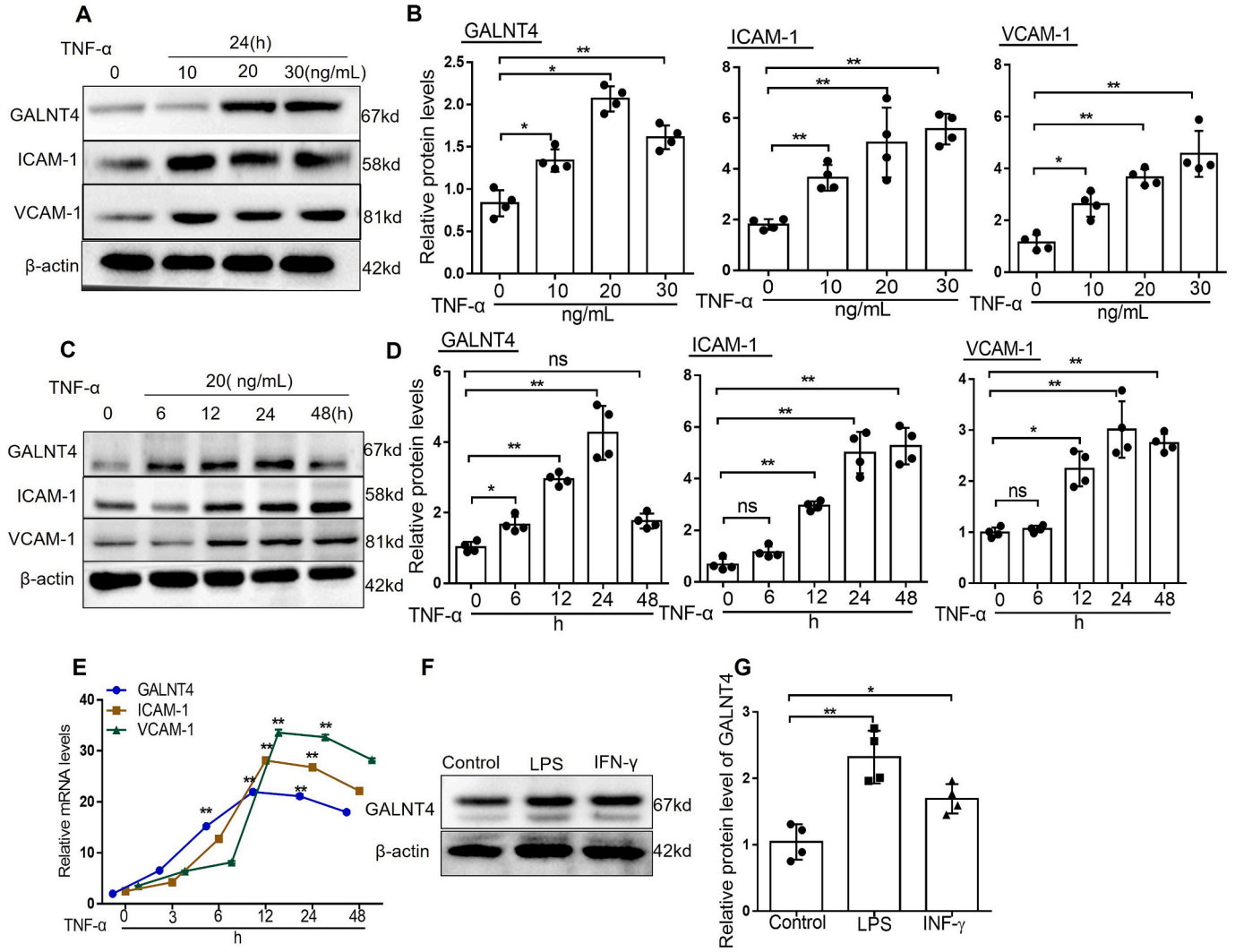


Fig. 3. GALNT4 was induced in HUVECs along with changes in the expression of adhesion molecules.
(A) HUVECs were treated with TNF-α (0, 10, 20, 30 ng/mL) for 24 h, and Western blot analysis of GALNT4, ICAM-1, and VCAM-1 expression. (B) Quantification of protein levels of GALNT4, ICAM-1, and VCAM-1(n = 4). (C) HUVECs were treated with TNF-α (20 ng/mL) for different times(0, 6, 12, 24, 48 h), and Western blot analysis of GALNT4, ICAM-1, and VCAM-1 expression. (D) Quantification of protein levels of GALNT4, ICAM-1, and VCAM-1(n = 4). (E) HUVECs were treated with TNF-α (20 ng/mL) for different times (0, 3, 6, 12, 24, 48 h), and the mRNA levels of GALNT4, ICAM-1, and VCAM-1 were detected by qRT-PCR(n = 3). (F and G) HUVECs were treated with LPS (10 ng/mL) or IFN-γ (10 ng/mL) for 24 h, and Western blot analysis of GALNT4 expression was performed (n = 4). *, $P < 0.05$, **, $P < 0.01$ vs control group, tested by ANOVA followed by Bonferroni post hoc tests.

lipopolysaccharide (LPS,10 ng/mL) and interferon-gamma (IFN-γ,10 ng/mL) also upregulated GALNT4 expression in HUVECs (Fig. 3F and G).

These data indicate that TNF-α upregulates GALNT4, similar to its induction of pro-inflammatory adhesion molecules, suggesting that GALNT4 may play a role in modulating the inflammatory response in ECs.

3.4. Knockdown of GALNT4 decreased monocyte-endothelial adhesion through downregulation of ICAM-1 and VCAM-1

To understand the role of GALNT4 in endothelial function, we next performed gain and loss-of-function studies in vitro. Firstly, GALNT4 was knocked down in HUVECs using specific siRNA. Results confirmed that this siRNA effectively reduced GALNT4 expression by approximately 81.6 % at both mRNA and protein levels (Fig. 4A to C). Using CM-Dil-labeled monocyte counting, we observed that GALNT4 knockdown significantly inhibited the adhesion of monocytes to ECs (Fig. 4D and E). Additionally, the protein levels of ICAM-1 and VCAM-1 were down-regulated by 48.9 ± 0.35 % and 38.5 ± 2.3 %, respectively, following GALNT4 knockdown (Fig. 4F and G).

These results confirmed that endogenous GALNT4 was required to maintain the excessive up-regulation of adhesion molecules in response to the pro-inflammatory stimuli.

3.5. Overexpression of GALNT4 promoted monocyte-endothelial adhesion mediated by upregulation of ICAM-1 and VCAM-1

Then, to investigate the role of GALNT4 in endothelial function, we further overexpressed GALNT4 in HUVECs. The efficiency of adenovirus-mediated overexpression was analyzed and confirmed by qRT-PCR (Fig. 5A) and Western blot (Fig. 5B and C). Our data indicated a significant increase in the number of monocytes adhering to the ECs in HUVECs overexpressing GALNT4 (Fig. 5D and E). Additionally, the overexpression of GALNT4 resulted in a marked increase in the expression of both ICAM-1 and VCAM-1(Fig. 5F and G).

These results suggested that enhanced GALNT4 expression promoted monocyte-endothelial adhesion mediated by upregulation of ICAM-1

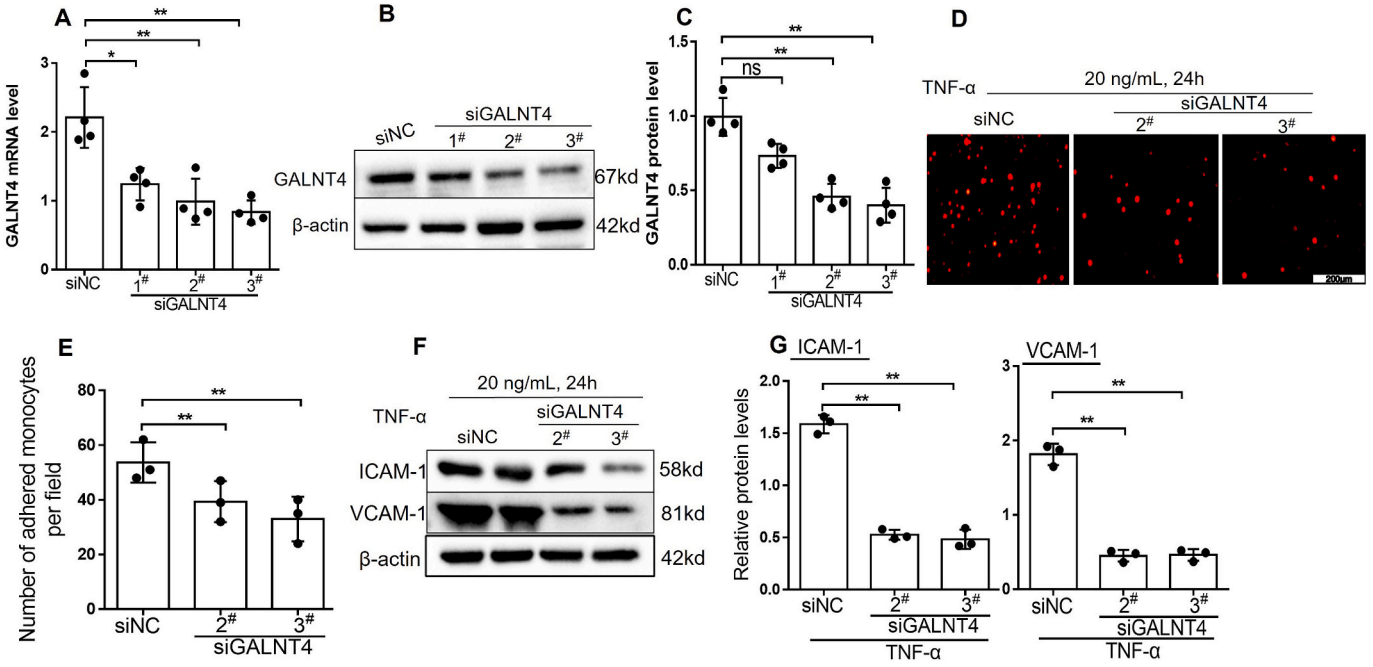


Fig. 4. Inhibition of GALNT4 decreased monocyte-endothelial adhesion by regulating the expression of ICAM-1 and VCAM-1.
(A) HUVECs were transfected with siRNA-negative control(siNC) or siRNA-GALNT4 (siGALNT4) (100 nM) for 48 h. The mRNA level of GALNT4 was detected by qRT-PCR (n = 4). (B) HUVECs were transfected with siNC or siGALNT4 for 48 h. The protein level of GALNT4 was detected by Western blot. (C) Quantification of protein levels of GALNT4 (n = 4). (D and E) HUVECs were transfected with siNC or siGALNT4 for 24 h, and then treated with TNF-α (20 ng/mL) for 24 h. Activated HUVECs were incubated with CM-Dil-labeled THP-1 cells for 30 min. D, Representative images of the adhesion of THP-1 to EC were visualized with a fluorescence microscope. Scale bar: 200 μm. E, The number of labeled THP-1 cells was counted from 6 random fields. (F) HUVECs were transfected with siNC or siGALNT4 for 24 h, and then treated with TNF-α (20 ng/mL) for 24 h. The protein level of ICAM-1 and VCAM-1 was detected by Western blot. (G) Quantification of protein levels of ICAM-1 and VCAM-1($n = 3$). *, $P < 0.05$, **, $P < 0.01$ vs siNC group, tested by ANOVA followed by Bonferroni post hoc tests.

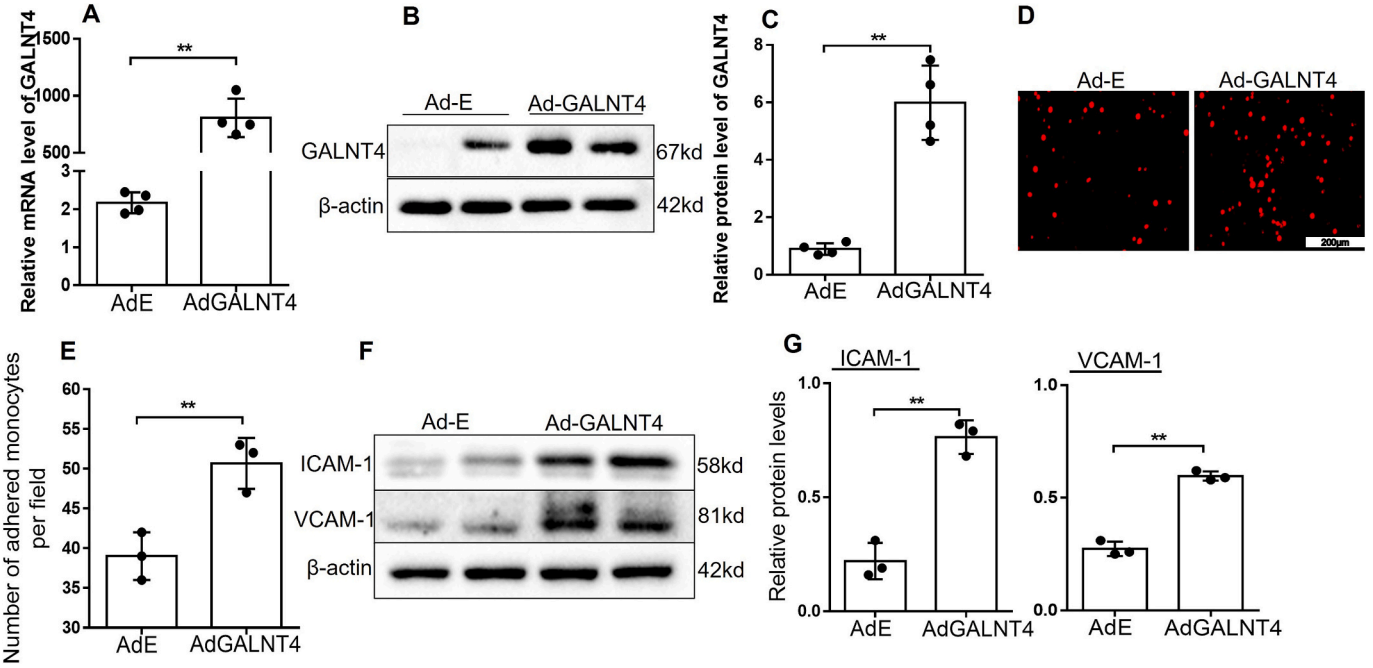


Fig. 5. Overexpression of GALNT4 promoted monocyte-endothelial adhesion by upregulating the expression of ICAM-1 and VCAM-1.
(A to C) HUVECs were infected with AdGALNT4 or empty vector (AdE) for 48 h, and total RNA and protein were extracted for the detection of GALNT4 by qRT-PCR (A) and Western blot (B and C). (D and E) HUVECs were infected with AdGALNT4 or AdE for 48 h. Activated HUVECs were incubated with CM-Dil-labeled THP-1 cells for 30 min. D, Representative images of the adhesion of THP-1 to EC were visualized with a fluorescence microscope. Scale bar: 200 μm. E, The number of labeled THP-1 cells was counted from 6 random fields. **, $P < 0.01$ vs AdE group, tested by two-tailed Student's *t*-test. (F and G) HUVECs were infected with AdGALNT4 or AdE for 48 h. The protein level of ICAM-1 and VCAM-1 was detected by Western blot. F, representative Western blots. G, Quantification of protein levels of ICAM-1 and VCAM-1($n = 3$). **, $P < 0.01$ vs AdE group, tested by two-tailed Student's t-test.

and VCAM-1.

3.6. GALNT4 regulated O-GalNAcylation of TNFR1 and the NF-κB signaling pathway

Given that GALNT4 modifies membrane proteins and that TNFR1 is the primary signaling receptor for our central inflammatory stimulus (TNF-α), we hypothesized that TNFR1, as a membrane protein, could be a target for GALNT4-mediated O-GalNAcylation. First, Lectin blot results in Fig. 6A revealed that GALNT4 knockdown reduced O-GalNAcylation levels. Subsequently, we performed immunoprecipitation using VVL (specific for GalNAc-O-Ser/Thr) agarose beads and probed the immunoprecipitated products with TNFR1 antibodies. Reciprocally, we immunoprecipitated HUVEC lysates with an anti-TNFR1 antibody and analyzed the enriched products using biotinylated VVL. These immunoprecipitation and lectin pull-down assays demonstrated that GALNT4 knockdown diminished TNFR1 O-GalNAcylation without affecting total TNFR1 protein levels, as indicated by reduced VVL pulldown efficiency (Fig. 6B and C). Conversely, GALNT4 overexpression enhanced TNFR1 glycosylation (Supplementary Fig. S4).

Furthermore, our findings indicated that TNF-α treatment elevated phosphorylated p65 (p-p65) levels in HUVECs. However, GALNT4 knockdown significantly attenuated TNF-α-induced p65 phosphorylation (Fig. 6D and E). Consistent with the in vitro findings, the attenuation of atherosclerosis by GALNT4 knockdown in *ApoE*$^{-/-}$ mice was associated with a significant reduction in the phosphorylation of NF-κB p65 in the aortic tissue, indicating a suppression of the NF-κB signaling pathway in vivo (Supplementary Fig. S5). In addition, IF staining for NF-κB translocation revealed that TNF-α promoted p65 nuclear translocation, an effect prevented by GALNT4 knockdown (Fig. 6F).

These findings suggest that GALNT4 regulates O-GalNAcylation of TNFR1 and the NF-κB signaling pathway.

3.7. GALNT4 regulated monocyte-endothelial adhesion and expression of adhesion molecules through the NF-κB signaling

To determine whether NF-κB participates in the GALNT4-mediated adhesion molecule and chemokine expression, we investigated the involvement of the NF-κB pathway. The effect of siRNA against p65 and the efficiency of the p65 siRNA were confirmed by qRT-PCR and Western blot (Supplementary Fig.S6). HUVECs were either pre-treated with the NF-κB inhibitor PDTC or transfected with p65 siRNA overnight, followed by infection with AdGALNT4. Both PDTC pretreatment and transfection significantly inhibited GALNT4-induced monocyte adhesion to ECs (Fig. 7A and B). Similarly, both interventions significantly reduced the upregulation of ICAM-1 and VCAM-1 caused by GALNT4 overexpression (Fig. 7C and D). To further validate NF-κB's role in adhesion molecule regulation, control adenovirus (AdE)-transduced HUVECs were treated with PDTC or p65 siRNA. Both interventions significantly attenuated TNF-α-induced monocyte adhesion (Supplementary Fig.S7), confirming that NF-κB activation is indispensable for ICAM-1/VCAM-1-dependent leukocyte recruitment.

In addition, we conducted a quantitative analysis of the NF-κB p65 binding activities in vitro by ChIP analysis. As shown in Fig. 7E, in vivo binding of NF-κB to the ICAM-1 and VCAM-1 promoter occurred after TNF-α stimulation. The binding activities of NF-κB were found to be

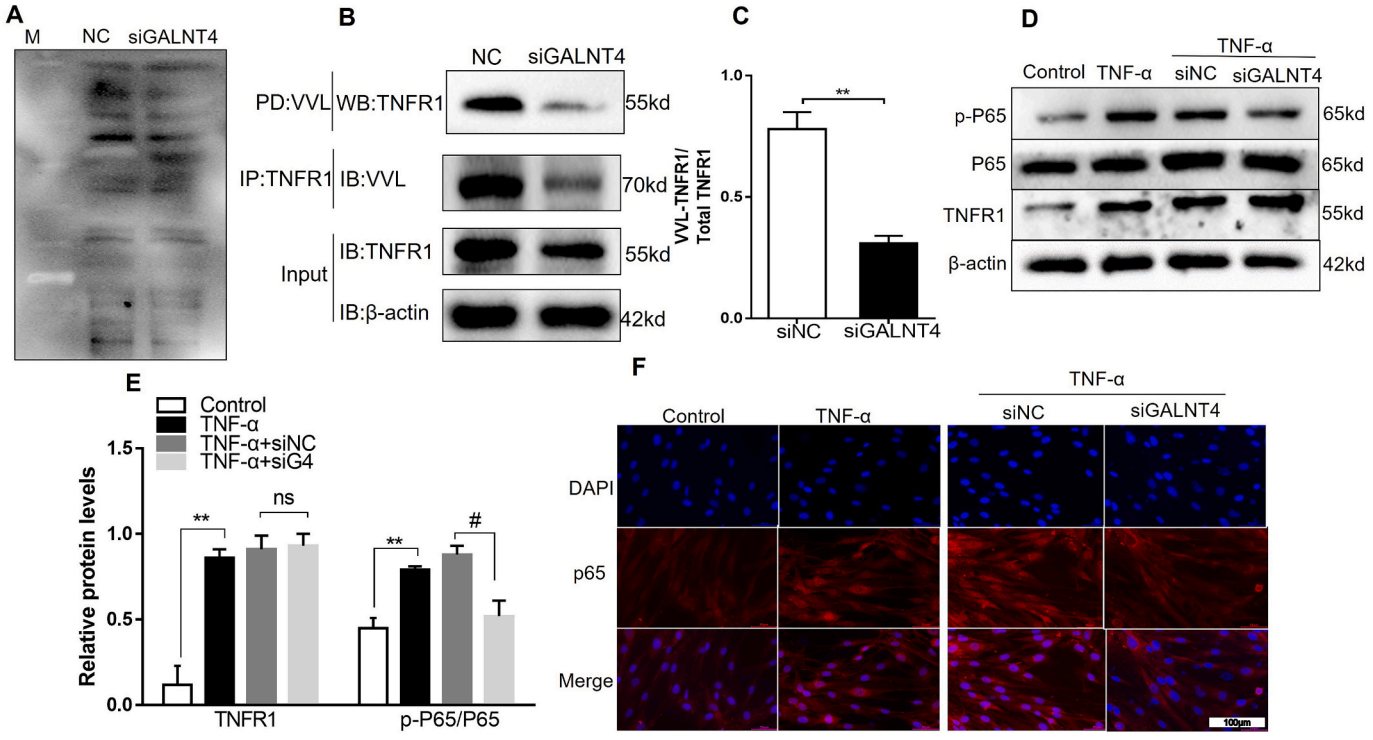


Fig. 6. GALNT4 regulated O-GalNAcylation of TNFR1 and the NF-κB signaling pathway.
(A) HUVECs were transfected with siNC or siGALNT4 for 48 h, and the level of total O-GalNAcylation was analyzed using Lectin blot. (B and C) HUVECs were transfected with siNC or siGALNT4 for 48 h. Lysates of HUVECs were immunoprecipitated with TNFR1 antibody or pulled down by VVL, then immunoblotted with VVL or TNFR1 antibody, respectively. B, representative IP blots; C, results were quantified from the pixel values in grayscales(n = 3). $^{**}P < 0.01$ versus siNC group, tested by the two-tailed Student's t-test. (D and E) HUVECs were transfected with siNC or siGALNT4 for 24 h, and then treated with TNF-α (20 ng/mL) for 24 h. Protein levels of TNFR1, P65, and p-P65 were detected by Western blot. D, representative Western blots; E, results were quantified from the pixel values in grayscales (n = 3). The fold change of p-P65 expression was normalized to total P65. ns, indicates no significance. $^{**}P < 0.01$ versus Control group. $\#\# P < 0.01$ versus siNC group, tested by one-way ANOVA and Tukey's multiple comparison test. (F) HUVECs were transfected with siNC or siGALNT4 for 24 h, and then treated with TNF-α (20 ng/mL) for 24 h or left untreated. Representative immunofluorescent staining against p65 (red) showed the p65 intranuclear translocation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

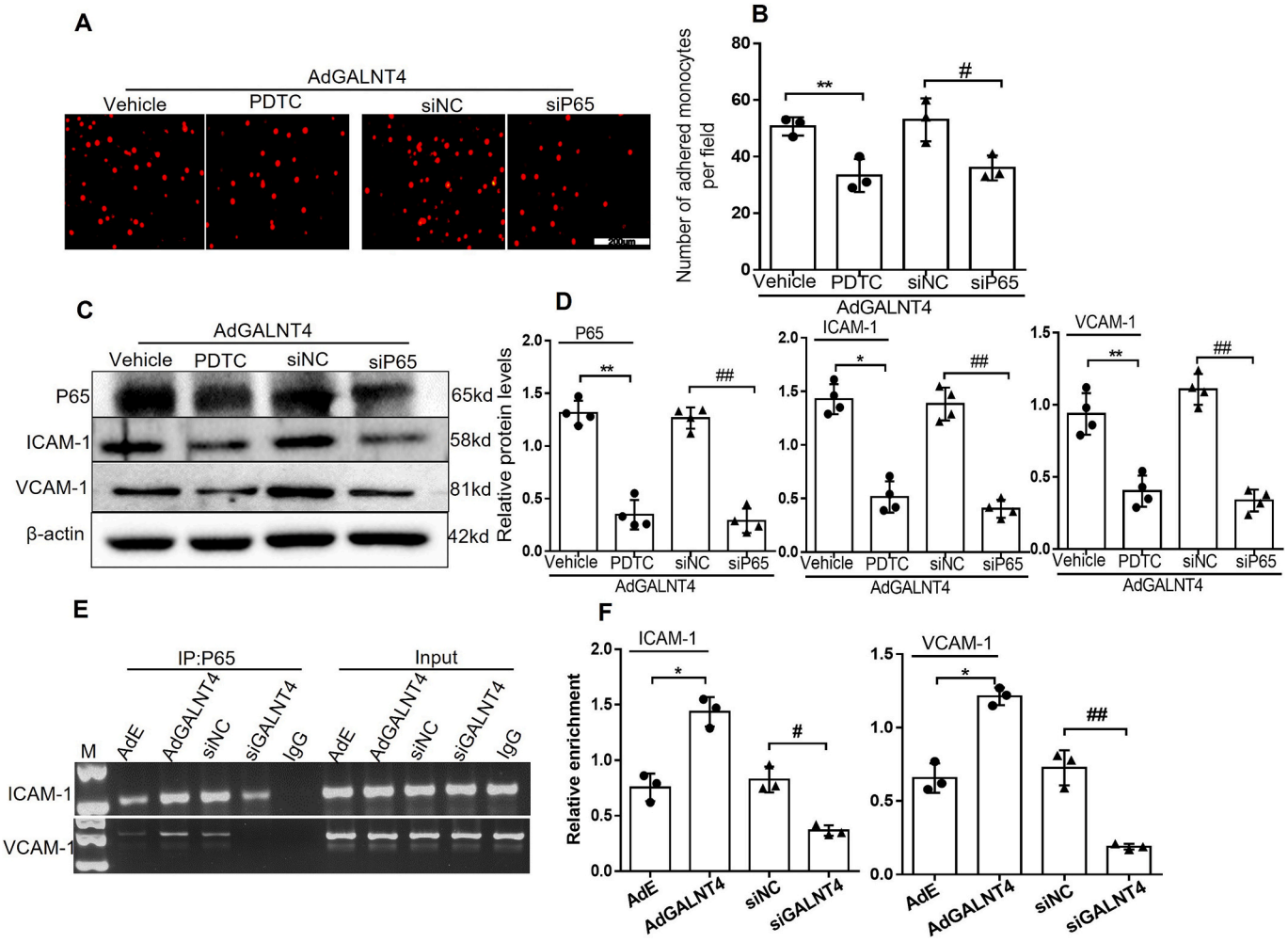


Fig. 7. GALNT4 regulated monocyte-endothelial adhesion and expression of adhesion molecules through the NF-κB signaling.
(A and B) HUVECs were pretreated with PDTC for 2 h or transfected with siNC or p65 siRNA(siP65) for 24 h, followed by infection with AdGALNT4 for another 24 h. Activated HUVECs were incubated with CM-Dil-labeled THP-1 cells for 30 min. A, Representative images of the adhesion of THP-1 to ECs were visualized with a fluorescence microscope. Scale bar: 200 μm. B, The number of labeled THP-1 cells was counted from 6 random fields. **, $P < 0.01$ vs vehicle-treated group, #, $P < 0.05$ vs siNC group, tested by two-tailed Student's t-test. (C and D) HUVECs were pretreated with PDTC for 2 h or transfected with siNC or p65 siRNA(siP65) for 24 h, followed by infection with AdGALNT4 for another 24 h. The protein level of ICAM-1 and VCAM-1 was detected by Western blot. C, representative Western blots; D, results were quantified from the pixel values in grayscales (n = 3). (E and F) HUVECs were transfected with siNC and siGALNT4, or infected with AdGALNT4 and AdE for 48 h, then stimulated with TNF-α (20 ng/mL) for 12 h. ChIP assays were performed with an antibody against p65 and normal rabbit IgG. The binding of p65 to the ICAM-1 and VCAM-1 promoters was determined with PCR. *, $P < 0.05$, vs AdE group, #, $P < 0.05$, ##, $P < 0.01$ vs siNC group, tested by two-tailed Student's t-test.

enhanced with overexpression of GALNT4, while they decreased with GALNT4 knockdown (Fig. 7E and F).

Taken together, NF-κB signaling pathways were involved in GALNT4-mediated monocyte-endothelial adhesion and expression of adhesion molecules.

4. Discussion

Our study demonstrates that GALNT4 expression is significantly elevated in human and mouse atherosclerotic plaques. GALNT4 knockdown attenuated atherosclerotic lesion development and endothelial inflammation in *ApoE*$^{-/-}$ mice. Mechanistically, GALNT4 promotes TNF-α-induced monocyte adhesion to ECs through glycosylation-dependent activation of NF-κB signaling and the subsequent upregulation of ICAM-1 and VCAM-1 expression.

GalNAc-type O-glycosylation is an abundant and highly diverse modification of proteins [17–19]. Initiated in the Golgi apparatus, this process is catalyzed by a large family of polypeptide *N*-acetylgalactosaminyltransferase (GalNAc-T, EC 2.4.1.41) isoenzymes, which transfer GalNAc to Ser, Thr, and possibly Tyr residues [20]. Currently, 20 human *GALNT* gene entries have been recorded [21,22], with *GALNT4* encoding the fourth member of this enzyme family. Emerging evidence suggests crucial roles for GALNT4 in the cardiovascular system [11,23,24]; however, its biological functions remain incompletely understood. Previous studies reported upregulated GALNT4 expression in diseased ECs, intimal lesions, and SMCs of *ApoE*$^{-/-}$ mice, contrasting with downregulation in foam cells, indicating its multi-cellular involvement in AS [10]. To clarify the role of GALNT4 in the initiation of AS, we localized the GALNT4 expression in the arteries. Consistent with prior findings, we observed significantly increased GALNT4 expression specifically in the ECs of atherosclerotic plaques. Although AS compromises endothelial integrity and reduces EC numbers, surviving cells adopt an activated, pro-inflammatory phenotype. The observed GALNT4 upregulation in plaque CD31^{+} cells is consistent with this pattern, reflecting increased expression per dysfunctional EC, not an increase in cell number. This GALNT4 induction may thus contribute to the inflammatory endothelial response that initiates AS. Furthermore, in vivo knockdown of GALNT4 attenuated atherosclerotic lesion size in *ApoE*$^{-/-}$ mice. The current findings confirm that expression of GALNT4 is promoted by the initiation of AS and provide evidence that GALNT4 may

participate in the pathogenesis of AS by regulating endothelial dysfunction.

Endothelial activation and inflammation are crucial for the initiation of AS [25]. Within this process, ICAM-1 and VCAM-1 play significant roles in mediating inflammation and facilitating leukocyte migration [4]. GALNT4 has been reported to play an important role in the interactions between ECs and monocytes by O-glycosylating the Thr residues of PSGL-1, which enables its interaction with P-selectin expressed on the surface of ECs. This interaction is critical for regulating monocyte tethering and rolling on ECs [11,26]. However, the specific role of GALNT4 within ECs remained unstudied before this work. Our current study demonstrated that GALNT4 expression rapidly increased in HUVECs following TNF-α treatment, mirroring the induction pattern of pro-inflammatory adhesion molecules. Furthermore, GALNT4 knockdown significantly inhibited the expression of ICAM-1 and VCAM-1 in ECs, both in vivo and in vitro. Therefore, inhibiting GALNT4 may prevent leukocyte adhesion to ECs, potentially providing therapeutic benefits in mitigating AS progression. This finding is consistent with the well-established central role of ICAM-1 and VCAM-1 as critical endothelial adhesion molecules governing firm monocyte adhesion under inflammatory conditions [4]. Future studies employing adhesion molecule-specific blocking antibodies or siRNA are warranted to formally establish their definitive causal role in this process.

TNF-α, a critical regulator of gene expression, induces the activation of the central transcription factor NF-κB [27]. Its receptor, TNFR1, is a membrane-spanning hub that integrates a multitude of downstream signaling cascades and is itself subject to glycosylation [14]. Here, we show that GALNT4 regulates O-GalNAcylation of TNFR1, thereby modulating TNF-α-induced NF-κB activation in HUVECs. This finding aligns with emerging evidence that O-GalNAc glycosylation fine-tunes receptor signaling in inflammation and disease [28,29]. TNFR1 forms pre-ligand clusters on the plasma membrane, and its oligomerization (trimers/hexamers) upon TNF-α binding is critical for initiating NF-κB signaling [30]. Reduced O-GalNAcylation (e.g., via GALNT4 knockdown) may impair TNFR1 signaling efficiency, as shown by diminished p65 phosphorylation and nuclear translocation. This parallels studies in which the sialylation or N-glycosylation of TNFR1 altered its stability and ligand binding [31]. Prior work has linked GALNT4 to PSGL-1 O-glycosylation in monocytes, promoting β2-integrin activation and adhesion via the Akt/mTOR/NF-κB axis [11]. Our data extend this paradigm to TNFR1, suggesting GALNT4 broadly regulates O-glycosylation-dependent inflammatory receptors. A key remaining question is the identification of the specific Thr or Ser residues on TNFR1 that are O-GalNAcylated by GALNT4. While our lectin pull-down and co-immunoprecipitation data confirm the interaction and modification, the definitive mapping of the glycosylation sites via mass spectrometry will be the focus of a subsequent, dedicated investigation.

Studies have demonstrated that blocking the NF-κB pathway effectively reduced plaque area and enhanced plaque stability in mice [16,32]. The current study reveals that GALNT4 regulates NF-κB-dependent inflammation in ECs. Specifically, GALNT4 knockdown attenuated TNF-α-induced phosphorylation of NF-κB p65. Moreover, both PDTC treatment and p65 siRNA transfection reduced monocyte adhesion to ECs induced by GALNT4. Additionally, GALNT4 knockdown attenuated TNF-α-induced activation of the ICAM-1 and VCAM-1 promoters. These findings support the hypothesis that GALNT4 promotes endothelial inflammation by activating the NF-κB signaling pathway, leading to increased transcription of pro-inflammatory genes. Our data place GALNT4 as an upstream regulator of this canonical pathway. It is well-established that anti-inflammatory cytokines offer protection against AS, and monoclonal antibodies targeting these cytokines have significantly transformed AS treatment [33,34]. Our findings identify GALNT4 as a novel regulator of endothelial inflammation and a potential therapeutic target for AS, as it positively regulates the expression of cell adhesion molecules. Consequently, targeting GALNT4 reduces the inflammatory response in ECs, thereby lowering the risk of AS.

A limitation of this study is that systemic AAV-shGALNT4 delivery may partially suppress GALNT4 in non-endothelial cells. Nevertheless, this finding is consistent with our comprehensive in vitro data using HUVECs, which establish a cell-autonomous role for GALNT4 in regulating the TNFR1/NF-κB axis in ECs. Future studies employing cell-type-specific knockout models will further dissect the contributions of GALNT4 in monocytes, SMCs, and other cell types.

In conclusion, this study demonstrates a novel role for GALNT4 in ECs, where it facilitates monocyte adhesion by activating NF-κB signaling in a glycosylation-dependent manner. Understanding the specific function and mechanism of GALNT4 in ECs could offer novel strategies for treating or preventing AS.

CRediT authorship contribution statement

Pengcheng Wei: Writing – original draft, Methodology. **Duan Li:** Software, Methodology. **Wenxuan Chen:** Methodology. **Xiaoxuan Zhang:** Software. **Wang Xinru:** Resources. **Zhao Han:** Methodology. **Yikai Wang:** Resources. **Liwei Guo:** Writing – review & editing, Funding acquisition.

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Declaration of competing interest

The manuscript has been read and approved by all authors for publication. It has not been published or is not being considered for publication elsewhere, in whole or part, in any language.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cellsig.2025.112209.

Data availability

No data was used for the research described in the article.

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